

6. Using the NLS panel data on N=716 young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10		Estimation Results for Exercise 15.6			
	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	−0.0033 (0.0011)	−0.0031 (0.0011)	−0.0012 (0.0011)	−0.0012 (0.0011)	−0.0023 (0.0007)
<i>SOUTH</i>	−0.2128 (0.0338)	−0.2384 (0.0344)	−0.3261 (0.1258)	−0.3261 (0.2495)	−0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

a. The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?

The OLS estimates for 1987 and 1988 are highly consistent. The numerical differences between the coefficients in column (1) and (2) are small, typically within one standard error of each other. This suggests that the structural relationship between the independent variables and $\ln(WAGE)$ remained stable over these two years.

In these individual-year OLS estimations, we are assuming intercept homogeneity. Specifically, we assume that there is no unobserved individual-specific effect correlated with the explanatory variables. By using OLS, we treat the data as a single cross-section for each year, assuming that the population parameters (including the intercept) are identical across all individuals in the sample.

b. The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it})$$

Explain any differences in assumptions between this model and the models in part (a).

According to the model, the error term is decomposed into two components. One is the individual-specific effect (u_i), which represents unobserved heterogeneity that is time-invariant. The other is the idiosyncratic error (e_{it}), which varies across both individuals and time.

The key difference in assumptions is: Unlike the OLS models in part (a) which lump everything into a single error term, this panel model acknowledges that u_i exists and stays constant for each individual, allowing us to control for person-specific traits that OLS might ignore.

c. Column (3) contains the estimated fixed effects model specified in part(b) .Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

Apart from the intercepts, the coefficient for *EXPER* shows the most significant difference. In the OLS estimations (Columns 1 and 2), the effect of experience is approximately 0.127, whereas in the Fixed Effects (FE) model (Column 3), it drops substantially to 0.0575.

The coefficient for *UNION* also shows a notable decrease from 0.144 to 0.0822. These differences suggest that the OLS estimates were biased upwards because they captured unobserved individual heterogeneity (such as ability or motivation) that is positively correlated with experience and union status. The FE model controls for these time-invariant traits, providing a more accurate estimate of the ceteris paribus effect.

d. The F-statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F-distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

$$N = 716, T = 2, K_s = 4$$

Suppose the F -statistic value is 11.68 and null hypothesis is consider as true, at 1% level of the significance, the degree of freedom is $N - 1 = 716 - 1 = 715$. Denominator degree of freedom is $NT - N - K_s = 1432 - 716 - 4 = 712$. $F_{0.99,715,712} = 1.190496$. The critical value at the 1% significance level is 1.190496. Reject region is $F > F_{0.99,715,712} = 1.190496$, $F = 11.68$, reject null hypothesis.

This indicates that there is significant unobserved individual heterogeneity among the individuals in the sample. Therefore, using a model that accounts for these individual differences is more appropriate than Pooled OLS.

e. Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?

1. Different Assumptions

- Column (3): Assumes idiosyncratic errors (e_{it}) are i.i.d., meaning there is no correlation between an individual's errors across different years.
- Column (4): Uses Cluster-robust standard errors, which allow for correlation within the same individual's errors over time (serial correlation), which is more realistic for panel data.

2. Comparison of Standard Errors

- Substantial Difference: The standard error for *SOUTH* shows the most significant change, nearly doubling from 0.1258 in column (3) to 0.2495 in column (4).
- Larger or Smaller: The robust standard errors are generally larger.

The increase in standard errors indicates that ignoring the correlation within individuals leads to an underestimation of the true variability. This is particularly evident for time-persistent variables like *SOUTH*,

where the traditional OLS/FE standard errors overstate the precision of the estimate.

f. Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

1. Coefficient Comparison: Apart from the intercept, the *EXPER* coefficient shows the most substantial difference, changing from 0.0575 in the FE model (3) to 0.0986 in the RE model (5).

2. Hausman Test: We compare RE with column (3) because the standard Hausman test assumes the idiosyncratic errors are i.i.d., which matches the assumptions used in the traditional FE model.

Based on the significant shift in coefficients (especially for *EXPER*), the Hausman test would likely reject the null hypothesis. This suggests that the unobserved individual heterogeneity (u_i) is correlated with the explanatory variables. Therefore, Random Effects is not appropriate for this model, and the Fixed Effects (FE) estimator should be preferred as it remains consistent.